

Strategic Recommendation Report

Hotel Booking Pricing Analysis

Executive Summary

This project analyzed hotel booking data to identify key factors influencing pricing, booking behavior, and customer cancellations. Through data cleaning, exploratory data analysis, statistical testing, customer segmentation, and dashboard development, several actionable insights were identified to support revenue optimization and customer retention. The following recommendations are designed to help the Revenue Management team improve profitability, occupancy, and operational efficiency.

Key Findings

1. Lead Time Significantly Impacts Cancellations

Statistical analysis revealed a strong relationship between booking lead time and cancellation behavior. Guests who book far in advance are more likely to cancel compared to those making last-minute reservations.

2. ADR Varies Across Customer Segments

Average Daily Rate (ADR) differs considerably based on market segment, customer type, room type, and seasonality. Certain customer groups consistently generate higher revenue than others.

3. Seasonal Demand Influences Revenue

Demand peaks during specific months, creating opportunities for revenue optimization through dynamic pricing strategies.

4. Repeat Guests Demonstrate Higher Loyalty

Repeat guests exhibit lower cancellation rates and provide more stable revenue streams compared to first-time customers.

5. Distribution Channels Affect Booking Performance

Some distribution channels contribute higher-value bookings and lower cancellation rates, indicating opportunities for channel optimization.

Strategic Recommendations

Recommendation 1: Implement Dynamic Pricing Strategies

Adjust room prices based on demand patterns, seasonality, booking lead time, and occupancy forecasts.

Expected Benefits:

- Increased revenue during peak demand periods.
- Improved occupancy during low-demand seasons.
- Better alignment between pricing and market conditions.

Actions:

- Increase ADR during high-demand months.
 - Offer promotional rates during off-peak periods.
 - Continuously monitor competitor pricing and market demand.
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Recommendation 2: Reduce Cancellation Rates

Develop targeted initiatives to minimize revenue loss caused by booking cancellations.

Expected Benefits:

- Improved revenue predictability.
- Higher occupancy rates.
- Reduced operational disruptions.

Actions:

- Introduce flexible rebooking options.
 - Encourage non-refundable bookings through discounts.
 - Send automated reminders before arrival dates.
 - Offer loyalty incentives for confirmed reservations.
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Recommendation 3: Strengthen Customer Retention Programs

Focus marketing and service initiatives on retaining high-value customers and increasing repeat bookings.

Expected Benefits:

- Higher customer lifetime value.
- Reduced acquisition costs.
- Improved brand loyalty.

Actions:

- Launch loyalty reward programs.
 - Provide exclusive offers for repeat guests.
 - Personalize communications and promotions based on booking history.
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Recommendation 4: Optimize Market Segment Targeting

Allocate marketing resources toward customer segments that generate higher ADR and lower cancellation rates.

Expected Benefits:

- Improved marketing ROI.
- Increased revenue per booking.
- Better customer acquisition efficiency.

Actions:

- Identify high-performing segments using dashboard insights.
 - Develop segment-specific promotional campaigns.
 - Reassess marketing spend across low-performing segments.
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Recommendation 5: Improve Revenue Management Monitoring

Utilize the Power BI dashboard as a continuous decision-support tool.

Expected Benefits:

- Faster identification of market trends.
- Data-driven pricing decisions.
- Enhanced visibility into operational performance.

Actions:

- Monitor ADR, occupancy, booking volume, and cancellations regularly.
 - Establish KPI thresholds and automated alerts.
 - Conduct monthly performance reviews using dashboard insights.
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Recommendation 6: Optimize Distribution Channel Performance

Evaluate booking channels based on revenue contribution and cancellation behavior.

Expected Benefits:

- Increased profitability.
- Lower distribution costs.
- Improved booking quality.

Actions:

- Prioritize high-performing channels.
 - Negotiate improved agreements with intermediaries.
 - Encourage direct bookings through website incentives.
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Expected Business Impact

By implementing these recommendations, the hotel can expect:

- Increased revenue through optimized pricing strategies.
 - Reduced cancellation-related losses.
 - Improved customer retention and loyalty.
 - Enhanced visibility into booking and pricing performance.
 - More effective allocation of marketing and operational resources.
 - Better support for data-driven revenue management decisions.
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Conclusion

The analysis demonstrates that pricing, customer behavior, seasonality, and booking patterns significantly influence hotel revenue performance. Leveraging these insights through dynamic pricing, customer retention initiatives, cancellation management strategies, and interactive dashboard monitoring can help the Revenue Management team maximize profitability while improving customer satisfaction and operational efficiency.